

Sociotechnical Safety Evaluation of Generative AI Systems

Key Takeaways

- The paper proposes a three-layered sociotechnical framework for AI safety evaluation consisting of capability evaluation (testing technical outputs in isolation), human interaction evaluation (assessing how context, users, and application design mediate harm), and systemic impact evaluation (measuring broader societal, economic, and environmental effects at scale).
- Current safety evaluations are overwhelmingly concentrated at the capability layer, meaning the field largely tests what AI systems can produce but rarely examines whether those outputs actually cause harm when real people use them in real contexts.
- The authors identify three critical evaluation gaps: a coverage gap (many risk areas like information hazards and human autonomy threats lack adequate evaluations), a context gap (human interaction and systemic impact evaluations are rare), and a multimodal gap (approximately 75% of evaluations target text output only despite growing deployment of image, audio, and video systems).
- Risks in multimodal AI systems can be compositional, meaning harm emerges specifically from combining modalities — for example, an innocuous caption becomes harassment when paired with a particular image — and these compositional risks require entirely novel evaluation approaches.
- Even in risk areas with seemingly robust evaluation coverage, such as representation harms with 83 identified evaluations, closer inspection reveals narrow focus on a few demographic axes like binary gender and occupation, leaving dimensions such as disability, age, religion, and social class largely unassessed.
- The authors emphasize that evaluation is inherently value-laden and incomplete: normative choices pervade decisions about what risks to measure, how to operationalize them into metrics, and what constitutes acceptable AI system performance.
- The paper argues that safety evaluations must become standardized, routine, and consequential rather than optional and ad hoc, with clearly delineated roles and responsibilities across AI developers, policymakers, civil society, and academic researchers.
- Drawing on system safety engineering, the paper argues that a paradigm shift is needed from component-based testing (assessing individual parts in isolation) to systems thinking that accounts for emergent harms arising from interactions between technical, human, and societal factors.

Common Areas of Confusion

- The three layers may be mistakenly interpreted as sequential or hierarchical steps, but the authors explicitly state that evaluations at each layer can and should be run simultaneously and are not ordered by importance or chronology — they represent different targets of analysis, not a pipeline.
- Readers may conflate 'capability' with 'what the system can do in practice,' but in this framework capability evaluation specifically refers to testing technical components and outputs in isolation, divorced from the real-world context that determines whether a capability actually causes harm.
- The term 'sociotechnical' may confuse readers who associate it loosely with 'involving society and technology'; here it has a precise meaning rooted in decades of research asserting that AI systems

are joint human-machine systems where social context co-determines outcomes, not merely systems that have social consequences.

- The distinction between human interaction evaluation (Layer 2) and systemic impact evaluation (Layer 3) can blur because both involve people and society; the key difference is that Layer 2 focuses on direct effects on individuals using or exposed to the system, while Layer 3 targets emergent, aggregate effects on broader structures like labor markets, public knowledge ecosystems, or the environment.
- Readers may assume the multimodal gap is simply about extending text evaluations to images or audio, but the paper emphasizes that risks across modalities are qualitatively different and compositional — meaning harm can emerge from the combination of modalities in ways that single-modality evaluations cannot detect even if adapted.
- The paper's acknowledgment that evaluation is 'never value-neutral' may be misread as undermining the value of evaluation altogether, when the authors' actual point is that normative assumptions should be made transparent and deliberate rather than hidden, and that this awareness strengthens rather than weakens the evaluation enterprise.
- The inclusion of adversarial testing as an example that straddles both the capability and human interaction layers may confuse readers about where particular methods belong; the authors clarify that methods are not rigidly assigned to layers but can serve different evaluative purposes depending on whether the focus is on the system output or the human experience of probing it.